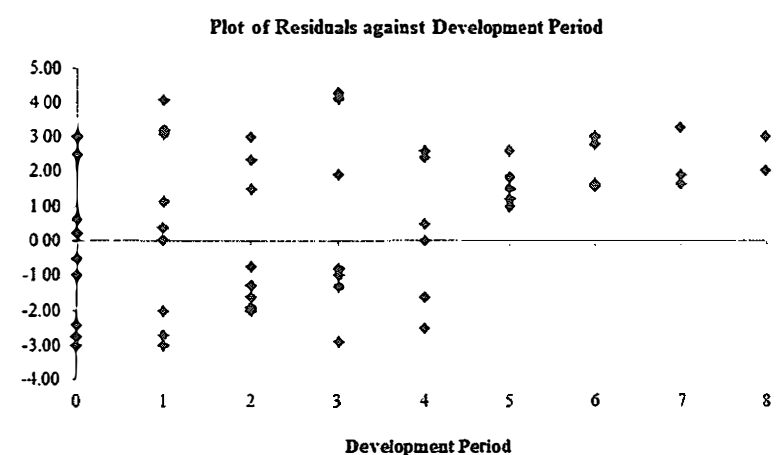
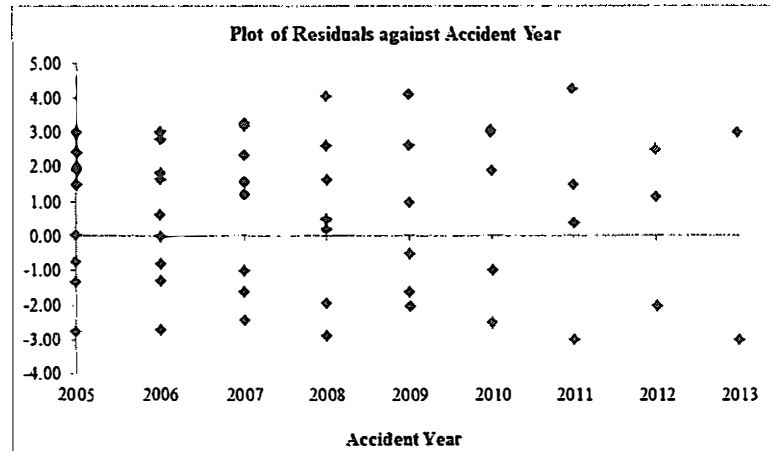


EXAM 7 – SPRING 2015

10. (2 points)

Given the following residual graphs from an over-dispersed Poisson (ODP) bootstrap model:



a. (0.5 point)

For each graph, briefly discuss whether the illustrated residuals demonstrate the presence or absence of heteroscedasticity.

b. (0.5 point)

Describe why it is important to adjust for heteroscedasticity when using a bootstrap model.

c. (1 point)

Describe two procedures for adjusting for heteroscedasticity.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 10	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: A7: Describe operational risk and demonstrate possible mitigation and quantification methodology.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> • No heteroscedasticity is present in the AY graph as the results appear random around 0 • Heteroscedasticity is present in the DP graph as there is a clear upward trend in the residuals starting with development period 5 <i>Sample Answer 2</i> • No heteroscedasticity is present in the AY graph as the residuals appear to have constant variance • Heteroscedasticity is present in the DP graph as the variance of the residuals by development period is different <i>Sample Answer 3</i> • No heteroscedasticity is present as the residuals appear to be random around 0, however, it's possible that it may exist with the more recent accident years and we simply don't know due to a low number of data points • Heteroscedasticity is present in the DP graph as the residuals in the later development periods are all above 0, which is not the case for the earlier development periods.	
Part b: 0.5 point	
<i>Sample Answer 1</i> Bootstrapping assumes residuals are independent and identically distributed. Heteroscedasticity violates this assumption as the residuals do not have constant variance. <i>Sample Answer 2</i> Bootstrap model samples residuals from all observed residuals to create new triangle from which to calculate LDFs. If residuals distributed differently in different accident years or development periods, it is not appropriate to sample from all residuals (assumption of i.i.d. residuals violated)	
Part c: 1 point	
<i>Sample Answer 1</i> • Stratified Sampling – Group residuals with like variances. Only sample residuals from these groups. • Hetero-Adjustment Factor – Group residuals with like variances. Calculate the standard deviation of each group. Adjust residuals with smaller variances upward by the ratio of the largest variance group to the group's variance. This allows us to sample from the entire triangle. After sampling, undo the adjustment to reflect the true relationship of the data. <i>Sample Answer 2</i> • Stratified Sampling – Group residuals together based on the size of their variance. For each part of the Triangles, sample only from the corresponding group of residuals where	

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

the sampled residuals and proceed with rest of the procedure.

- Hetero Adjustment – Group residuals based on size of variance for each group determine its standard deviation divide standard deviation by standard deviation of largest group. Multiply all residuals in group by that factor, then we can sample residuals, divide by factors before using to calculate pseudo triangles.

Other Acceptable Answer for the Hetero-Adjustment Factor procedure

- Adjustment: Adjust residuals by multiplying residuals of homogeneous groups by a constant factor [max standard deviation / standard deviation of the group] to give residuals homogeneous variance. Then, divide by the factor once residuals have been sampled.
- Make a heteroscedasticity adjustment to all the data to bring all the variances in line with each other. Run the bootstrapping process with the adjusted data then undo the adjustment to return the results to their original level once the process is complete.

EXAMINERS' REPORT

Candidates performed well on this question overall but part c was challenging. Candidates were expected to know what heteroscedasticity meant and how it applied to the graph shown. Candidates were also expected to know at a high level the adjustments that can be applied to correct for heteroscedasticity.

Part a

Candidates performed extremely well on this part. A few common mistakes were mixing up hetero- and homoscedasticity, and not including any justification for the presence/absence thereof.

Part b

Candidates did fairly well here. Most candidates identified the assumption of the bootstrap model that residuals are i.i.d., but a fair number of candidates did not sufficiently demonstrate how heteroscedasticity violates this assumption. Many candidates who received full credit didn't sufficiently define heteroscedasticity explicitly in this part, but had enough detail from part a to compensate for an otherwise insufficient answer here.

Part c

Candidates did well identifying and explaining the main points of stratified sampling; however, a common error was not giving enough detail regarding the hetero-adjustment factor approach. This is understandable, given that this method simply has more detail to it than stratified sampling. It is worth noting that candidates did not need to give any formulas to receive full credit if their answer contained all the high level aspects of the approach.